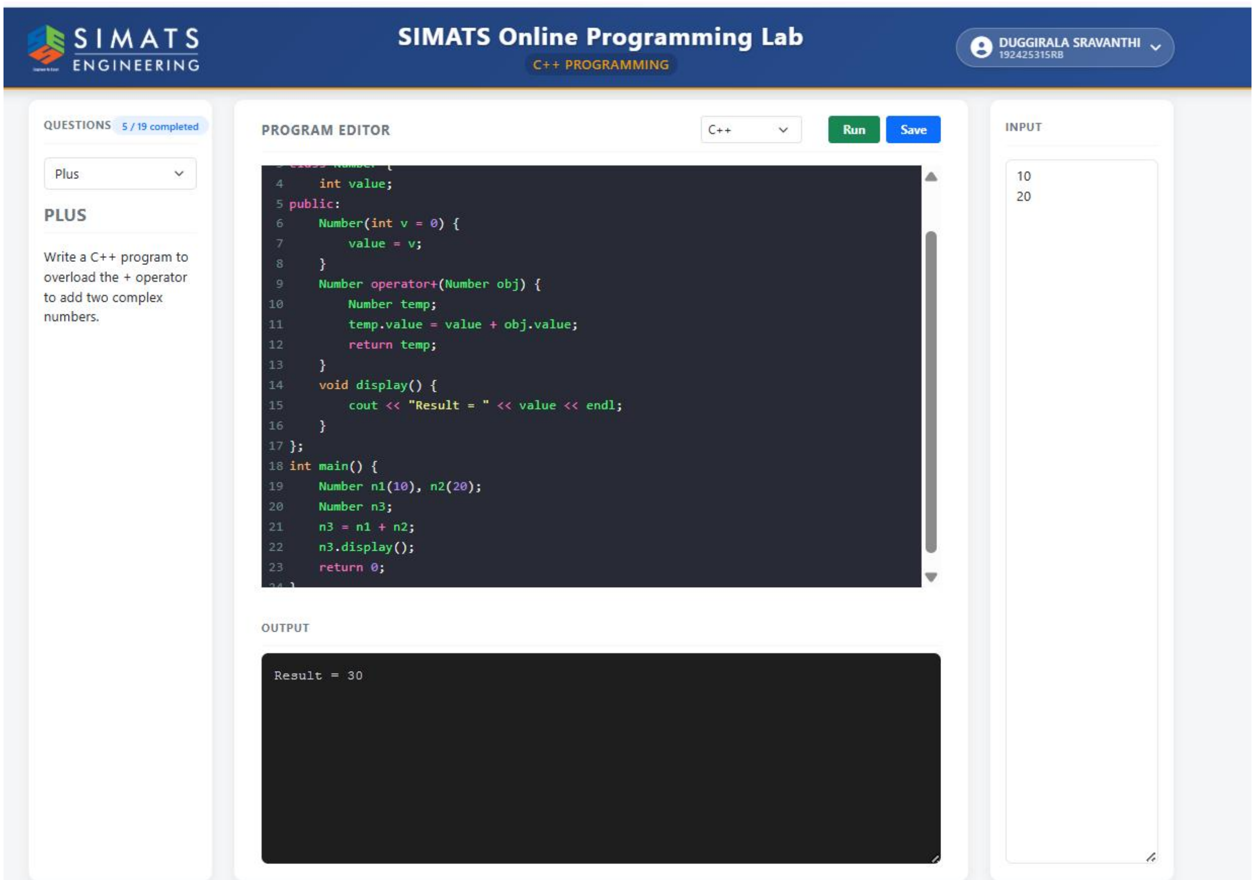
SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

DSA0115 Object Oriented Programming with C++ for System Level Programming

OPERATOR OVERLOADING



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C++ PROGRAMMING

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QUESTIONS 5 / 19 completed

Increment

INCREMENT

Write a C++ program to overload the ++ operator to increment a variable.

PROGRAM EDITOR

C++ Run Save

```
using namespace std;
class Number {
    int num;
public:
    void getData() {
        cout << "Enter a number: ";
        cin >> num;
    }
    void operator++() {
        ++num;
    }
    void display() {
        cout << "Value after increment = " << num << endl;
    }
};
int main() {
    Number n;
    n.getData();
    ++n;
    n.display();
    return 0;
```

OUTPUT

```
Enter a number: Value after increment = 16
```

INPUT

15

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<<

<<

Write a C++ program to overload the << operator to print the contents of a user-defined class.

PROGRAM EDITOR

C++ Run Save

```cpp
#include <iostream>
using namespace std;
class Student {
    int id;
    string name;
public:
    Student(int i, string n) {
        id = i;
        name = n;
    }
    friend ostream& operator<<(ostream &out, Student s) {
        out << "ID : " << s.id << endl;
        out << "Name : " << s.name << endl;
        return out;
    }
};
int main() {
    Student s(101, "Sravanthi");
    cout << s;
    return 0;
}
```

OUTPUT

```
ID : 101
Name : Sravanthi
```

INPUT

101
Sravanthi

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QUESTIONS 5 / 19 completed

Equal

EQUAL

Write a C++ program to overload the == operator to compare two objects of a user-defined class.

PROGRAM EDITOR

C++ Run Save

```cpp
#include <iostream>
using namespace std;
class Number {
    int value;
public:
    Number(int v = 0) {
        value = v;
    }
    bool operator==(Number obj) {
        return value == obj.value;
    }
};
int main() {
    Number n1(25), n2(25);
    if (n1 == n2)
        cout << "Both are Equal";
    else
        cout << "Not Equal";
    return 0;
}
```

OUTPUT

```
Both are Equal
```

INPUT

```
10
10
```

QUESTIONS 5 / 19 completed

[]

[]

Write a C++ program to overload the [] operator to access the elements of an array using an index.

PROGRAM EDITOR

C++

Run Save

```cpp
class Array {
    int arr[5];
public:
    void getData() {
        cout << "Enter 5 array elements:\n";
        for (int i = 0; i < 5; i++) {
            cin >> arr[i]; }}
    int operator[](int index) {
        if (index >= 0 && index < 5)
            return arr[index];
        else {
            cout << "Invalid Index!" << endl;
            return -1;} }};
int main() {
    Array a;
    int index;
    a.getData();
    cout << "Enter index to access (0-4): ";
    cin >> index;
    cout << "Element at index " << index << " = " << a[index] << endl;
    return 0;
```

OUTPUT

```
Enter 5 array elements:
Enter index to access (0-4): Element at index 2 = 30
```

INPUT

```
10
20
30
50
58
2
```